

Appendix A for “Multi-level regression and post-stratification for discrete choice modelling and stated preference research”

Region-level (i.e. Level-2) variables in the MRP model

In the Saskatchewan River Delta (SRD) empirical example, several of the provinces only have a handful of respondents given the quota-sampling strategy employed. For example, only 4 respondents were from Prince Edward Island. To facilitate the pooling of preference information, one can include additional parameters that help explain preference differences among provinces. For example, a region random parameter which groups provinces of Canada into four regions¹ can be included or provincial-level variables such as the average right-wing political party vote share in the 2021 election (X_{rw}).² The purpose of these ‘Level 2’ variables is to help inform the imputation of WTP in step 2.

The revised specification for Equation (9) is now

$$\omega_i = \beta_0 + \alpha_{a,g[i]}^{age-gender} + \alpha_{p[i]}^{province} + \alpha_{m[i]}^{income} + \alpha_{e[i]}^{education}$$

$$\alpha_{a,g}^{age-gender} \sim N(0, \sigma_{age-gender}^2), \text{ for } a = 1, \dots, 5, g = 1, 2 \quad (1)$$

$$\alpha_m^{income} \sim N(0, \sigma_{income}^2), \text{ for } m = 1, \dots, 7 \quad (2)$$

$$\alpha_e^{education} \sim N(0, \sigma_{education}^2), \text{ for } e = 1, \dots, 5 \quad (3)$$

$$\alpha_p^{province} \sim N(\alpha_{r[p]}^{region} + \beta_1 X_{rw}, \sigma_{province}^2), \text{ for } p = 1, \dots, 11 \quad (4)$$

$$\alpha_r^{region} \sim N(0, \sigma_{region}^2), \text{ for } r = 1, \dots, 4 \quad (5)$$

With only 11 provinces, there is little empirical benefit gained from including these Level-2 variables in the current SRD setting. However, for contexts where there are many more geographic regions to predict, such as the 50 American states, the appeal of using these Level-2 variables is higher.

¹The four regions include British Columbia and the 3 northern territories, the three prairie provinces (Alberta, Saskatchewan, and Manitoba), Ontario and Quebec, and Atlantic Canada (New Brunswick, Nova Scotia, Prince Edward Island, and Newfoundland and Labrador).

²The right wing vote share aggregates the vote share for the two main right wing parties in that election: Conservative Party of Canada and the People’s Party of Canada.

Additional Tables and Figures

Table A1: WTP-space logit model estimates for a provincial wetland restoration program across the 3 Prairie provinces

	Alberta	Saskatchewan	Manitoba	All 3 Priarie Provinces
Scale (λ)	0.150 (0.026)	0.236 (0.045)	0.157 (0.025)	0.165 (0.016)
Program constant	1.119 (0.158)	1.162 (0.179)	1.628 (0.112)	1.381 (0.077)
nObs	870	318	811	1999

The program constant can be interpreted as the WTP for a provincial wetland restoration program in hundreds of dollars because of the cost rescaling for estimation. Parameter diagnostic metrics for R-hat and effective sample size are provided in Table A5.

Table A2: WTP-space MRP model estimates for a provincial wetland restoration program across the 3 Prairie provinces

	Alberta	Saskatchewan	Manitoba	All 3 Priarie Provinces
Scale (λ)	0.146 (0.026)	0.229 (0.047)	0.157 (0.025)	0.168 (0.017)
Program constant	0.699 (0.459)	0.784 (0.485)	1.453 (0.308)	1.082 (0.388)
SD age-gender	0.334 (0.288)	0.308 (0.291)	0.321 (0.220)	0.169 (0.134)
SD education	0.392 (0.367)	0.358 (0.338)	0.386 (0.287)	0.324 (0.226)
SD income	0.623 (0.453)	0.679 (0.471)	0.236 (0.170)	0.383 (0.182)
SD province				0.472 (0.367)
nObs	870	318	811	1999

The program constant can be interpreted as the WTP for a provincial wetland restoration program in hundreds of dollars because of the cost rescaling for estimation. Parameter diagnostic metrics for R-hat and effective sample size are provided in Table A6.

Table A3: WTP-space MRP model estimates for a provincial wetland restoration program using the validity sample

	Alberta	Saskatchewan	Manitoba	All 3 Priarie Provinces
Scale (λ)	0.112 (0.030)	0.229 (0.064)	0.155 (0.036)	0.150 (0.022)
Program constant	-0.393 (0.764)	0.186 (0.704)	1.186 (0.450)	0.595 (0.572)
SD age-gender	0.627 (0.722)	0.500 (0.539)	0.611 (0.446)	0.317 (0.263)
SD education	0.857 (1.153)	0.591 (0.703)	0.277 (0.325)	0.235 (0.227)
SD income	0.816 (0.773)	0.713 (0.746)	0.260 (0.239)	0.337 (0.268)
SD province				0.896 (0.785)
nObs	526	192	459	1177

The program constant can be interpreted as the WTP for a provincial wetland restoration program in hundreds of dollars because of the cost rescaling for estimation. Parameter diagnostic metrics for R-hat and effective sample size are provided in Table A7.

Table A4: Diagnostics (Rhat, ESS) for SRD restoration program WTP-space logit models

Model	Term	Rhat	ESS(bulk)	ESS(tail)
MNL	Program constant	1.00	2568	2850
MNL	Scale (λ)	1.00	3338	2695
MNL	Mean Sturgeon	1.00	3364	2984
MNL	Mean Habitat	1.00	3565	3180
MNL	Mean Waterfowl	1.00	3734	3303
MNL	Mean Muskrat	1.00	3676	3182
MRP	Program constant	1.00	5802	2548
MRP	Scale (λ)	1.00	5130	3304
MRP	Mean Sturgeon	1.00	1770	1989
MRP	Mean Habitat	1.00	1594	1611
MRP	Mean Waterfowl	1.01	1107	1255
MRP	Mean Muskrat	1.00	1437	1517
MRP	SD Sturgeon: age-gender	1.00	1166	1501
MRP	SD Sturgeon: education	1.00	1223	1666
MRP	SD Sturgeon: income	1.00	1432	1714
MRP	SD Sturgeon: province	1.01	1136	1711
MRP	SD Habitat: age-gender	1.00	1427	1893
MRP	SD Habitat: education	1.00	1116	1501
MRP	SD Habitat: income	1.00	1293	1386
MRP	SD Habitat: province	1.00	1352	2196
MRP	SD Waterfowl: age-gender	1.00	1288	1594
MRP	SD Waterfowl: education	1.00	1313	1692
MRP	SD Waterfowl: income	1.00	2318	2447
MRP	SD Waterfowl: province	1.00	1575	2185
MRP	SD Muskrat: age-gender	1.00	1164	1305
MRP	SD Muskrat: education	1.00	1284	1521
MRP	SD Muskrat: income	1.00	1192	1615
MRP	SD Muskrat: province	1.00	1541	1737
RPL	Program constant	1.00	730	1241
RPL	Scale (λ)	1.00	1669	2691
RPL	Mean Sturgeon	1.00	2662	2834
RPL	Mean Habitat	1.00	3184	2848
RPL	Mean Waterfowl	1.00	3780	2704
RPL	Mean Muskrat	1.00	3397	3148
RPL	SD Program: individual	1.01	849	1800
RPL	SD Sturgeon: individual	1.00	775	1561
RPL	SD Habitat: individual	1.00	803	1484
RPL	SD Waterfowl: individual	1.01	347	896
RPL	SD Muskrat: individual	1.00	626	901

Table A5: Diagnostics (Rhat, ESS) for WTP-space wetland logit models

Model	Term	Rhat	ESS(bulk)	ESS(tail)
Alberta	Program constant	1	2356	1952
Alberta	Scale (λ)	1	2582	2131
Saskatchewan	Program constant	1	2003	1478
Saskatchewan	Scale (λ)	1	1798	1426
Manitoba	Program constant	1	1439	1924
Manitoba	Scale (λ)	1	1277	1425
All 3 Prairie Provinces	Program constant	1	2296	2343
All 3 Prairie Provinces	Scale (λ)	1	2272	2060

Note:

Rhat close to 1.00 indicates convergence; ESS values represent effective sample sizes for bulk and tail distributions.

Table A6: Diagnostics (Rhat, ESS) for WTP-space wetland MRP models

Model	Term	Rhat	ESS(bulk)	ESS(tail)
Alberta	Program constant	1	1912	1557
Alberta	SD age-gender	1	1635	1849
Alberta	SD education	1	1639	1932
Alberta	SD income	1	1092	1485
Alberta	Scale (λ)	1	3400	1807
Saskatchewan	Program constant	1	1793	1826
Saskatchewan	SD age-gender	1	1951	1842
Saskatchewan	SD education	1	1867	1684
Saskatchewan	SD income	1	1198	1096
Saskatchewan	Scale (λ)	1	3192	2336
Manitoba	Program constant	1	1818	2181
Manitoba	SD age-gender	1	1388	1698
Manitoba	SD education	1	1409	1461
Manitoba	SD income	1	1541	1982
Manitoba	Scale (λ)	1	4680	2614
All 3 Prairie Provinces	Program constant	1	1482	1572
All 3 Prairie Provinces	SD age-gender	1	1219	1446
All 3 Prairie Provinces	SD education	1	1238	1260
All 3 Prairie Provinces	SD income	1	1692	2419
All 3 Prairie Provinces	SD province	1	1283	1534
All 3 Prairie Provinces	Scale (λ)	1	4647	2711

Note:

Rhat close to 1.00 indicates convergence; ESS values represent effective sample sizes for bulk and tail distributions.

Table A7: Diagnostics (Rhat, ESS) for WTP-space wetland MRP models using the validity sample

Model	Term	Rhat	ESS(bulk)	ESS(tail)
Alberta	Program constant	1	3555	2713
Alberta	SD age-gender	1	2840	2383
Alberta	SD education	1	2262	2092
Alberta	SD income	1	1844	1776
Alberta	Scale (λ)	1	2936	1998
Saskatchewan	Program constant	1	3313	2671
Saskatchewan	SD age-gender	1	2610	2194
Saskatchewan	SD education	1	2211	2364
Saskatchewan	SD income	1	1766	2196
Saskatchewan	Scale (λ)	1	3234	2078
Manitoba	Program constant	1	2069	1635
Manitoba	SD age-gender	1	1274	765
Manitoba	SD education	1	1844	1652
Manitoba	SD income	1	2107	2238
Manitoba	Scale (λ)	1	3890	2054
All 3 Prairie Provinces	Program constant	1	1886	1408
All 3 Prairie Provinces	SD age-gender	1	1321	1367
All 3 Prairie Provinces	SD education	1	1906	1702
All 3 Prairie Provinces	SD income	1	1335	1735
All 3 Prairie Provinces	SD province	1	1403	1127
All 3 Prairie Provinces	Scale (λ)	1	2817	1838

Note:

Rhat close to 1.00 indicates convergence; ESS values represent effective sample sizes for bulk and tail distributions.

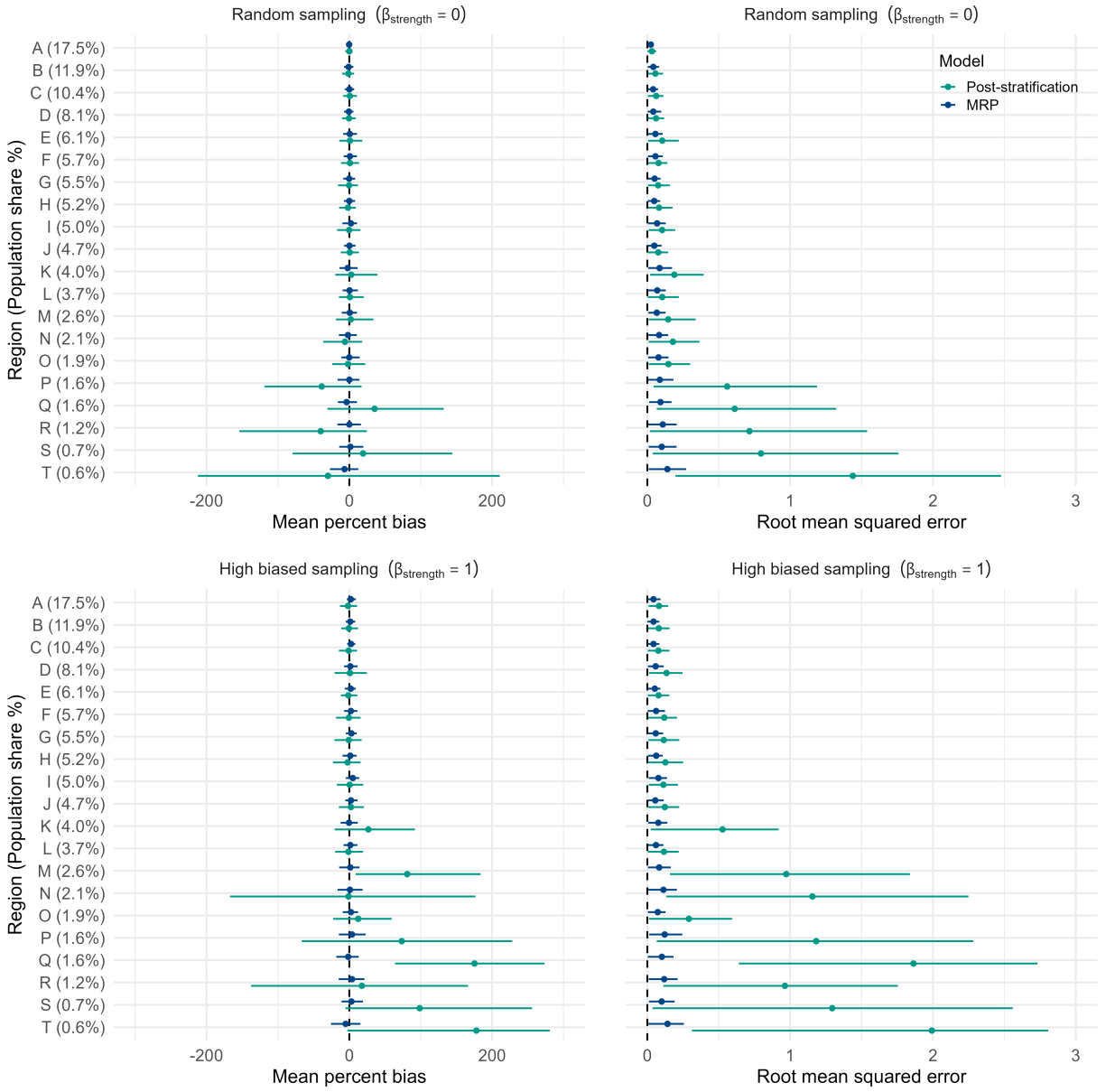


Figure A1: Comparison of MRP and post-stratification for regional WTP estimation with 3000 survey sample

If Alberta holds a provincial referendum to determine whether to put into place the Wetland Restoration Plan and you are asked to vote for or against the program, what would you choose?

Please read the following two statements and choose the one that indicates how you would vote.

☐ **I vote Yes.** I am willing to pay \$125 more in annual provincial income taxes for my household for 10 years starting in January 2022 to pay for the Wetland Restoration Plan that increases the amount of wetlands by 10% in the Prairie Pothole region of Alberta.

☐ **I vote No.** I am not willing to pay \$125 more in annual provincial income taxes for my household for 10 years starting in January 2022 to pay for the Wetland Restoration Plan that increases the amount of wetlands by 10% in the Prairie Pothole region of Alberta.

Figure A2: Example of single binary choice question for wetland survey

Alternatives A and B are potential Saskatchewan River Delta futures. The Status Quo alternative means no new conservation occurs. Given the choice between these three alternatives, how would you vote?

	Results in 20 years		
	Status Quo	Alternative A	Alternative B
Lake Sturgeon	30% 3,000 of 10,000 fish conservation target	15% 1,500 of 10,000 fish conservation target	100% 10,000 of 10,000 fish conservation target
Waterfowl population	25% 200,000 of 800,000 breeding ducks possible	50% 400,000 of 800,000 breeding ducks possible	75% 600,000 of 800,000 breeding ducks possible
Muskrat abundance	5% 1 muskrat found per hectare out of 20 possible	30% 6 muskrats found per hectare out of 20 possible	60% 12 muskrats found per hectare out of 20 possible
Habitat in healthy ecological condition	45% 405,000 of 900,000 hectares	60% 540,000 of 900,000 hectares	75% 675,000 of 900,000 hectares
Annual cost to your household for 20 years	\$0 Increase in annual taxes for 20 years	\$15 Increase in annual taxes for 20 years	\$325 Increase in annual taxes for 20 years
I would vote for...	☐ Status Quo	☐ Alternative A	☐ Alternative B

Figure A3: Example of a choiceset for the Saskatchewan River Delta (SRD) study

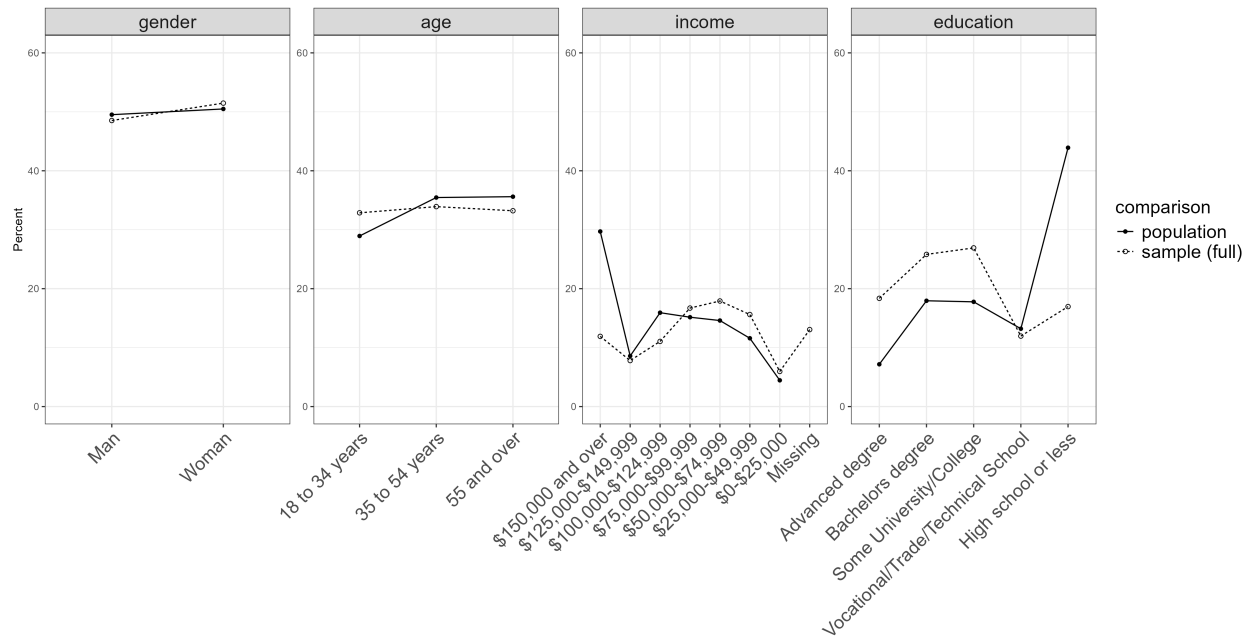


Figure A4: Comparison of wetland survey sample and target population across all three Priarie provinces

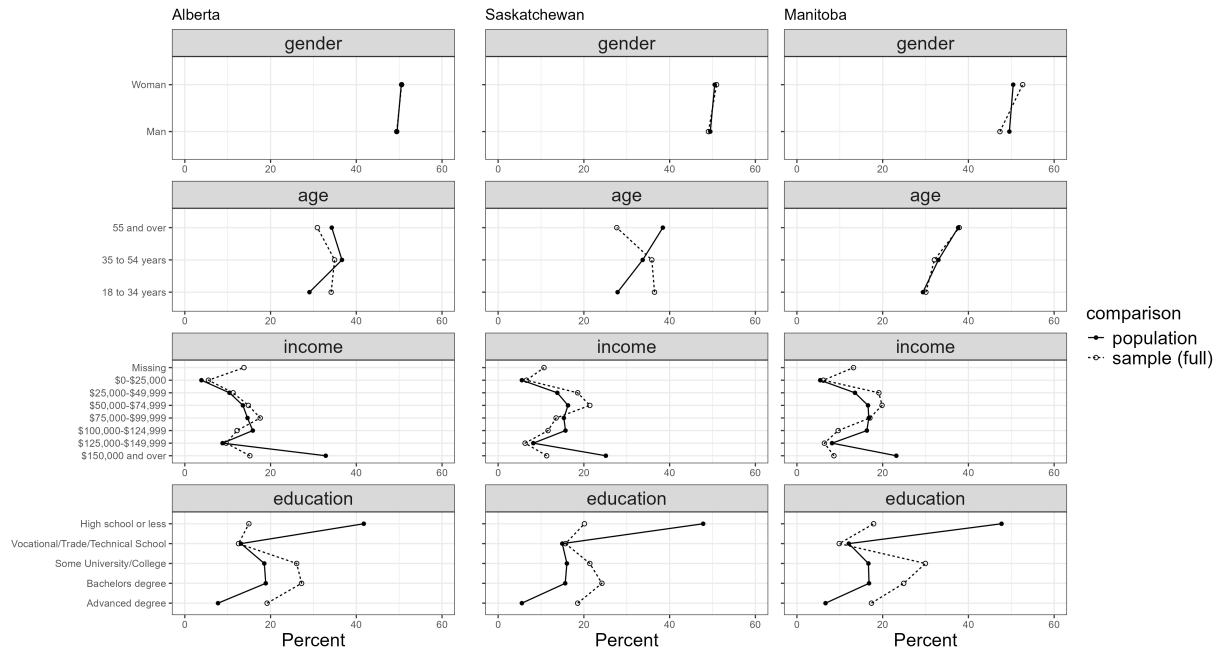


Figure A5: Comparison of wetland survey sample and target population for each province

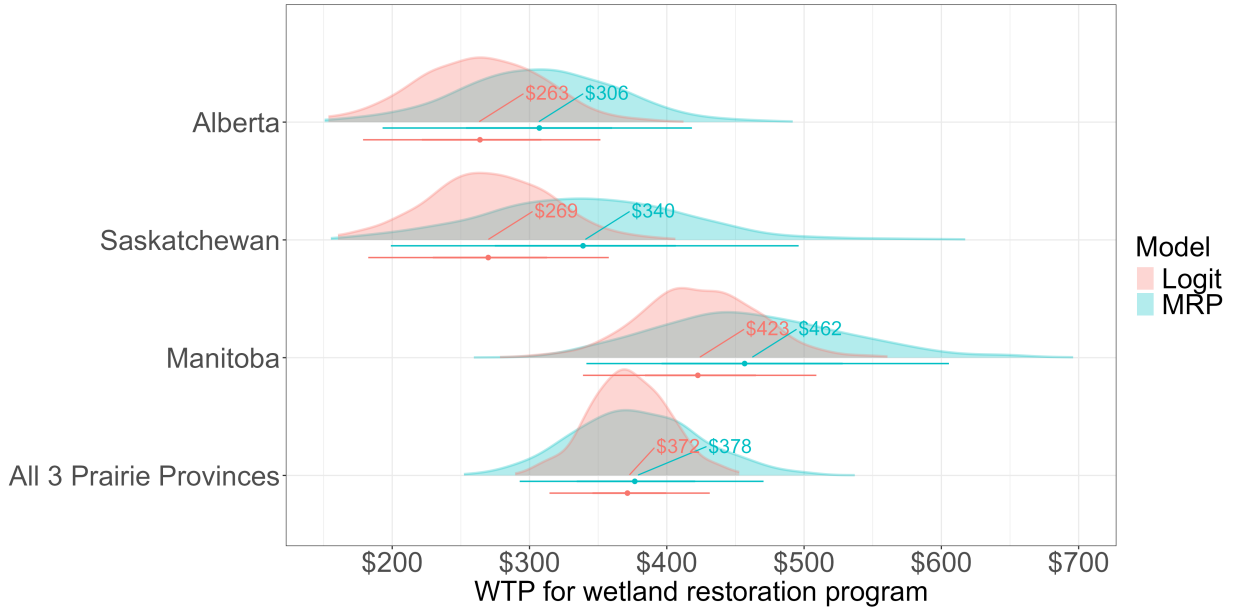


Figure A6: WTP for wetland restoration using alternative Logit and MRP specifications without exponentiation

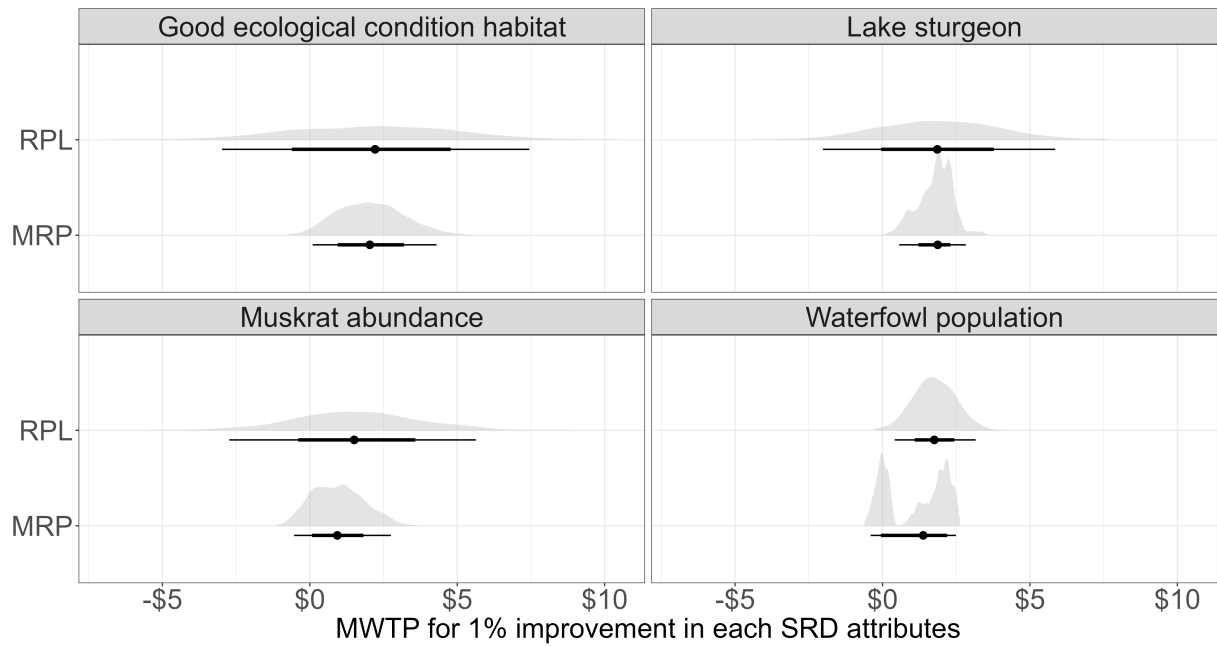


Figure A7: Distribution of MWTP for each SRD attribute using the MRP and RPL approach

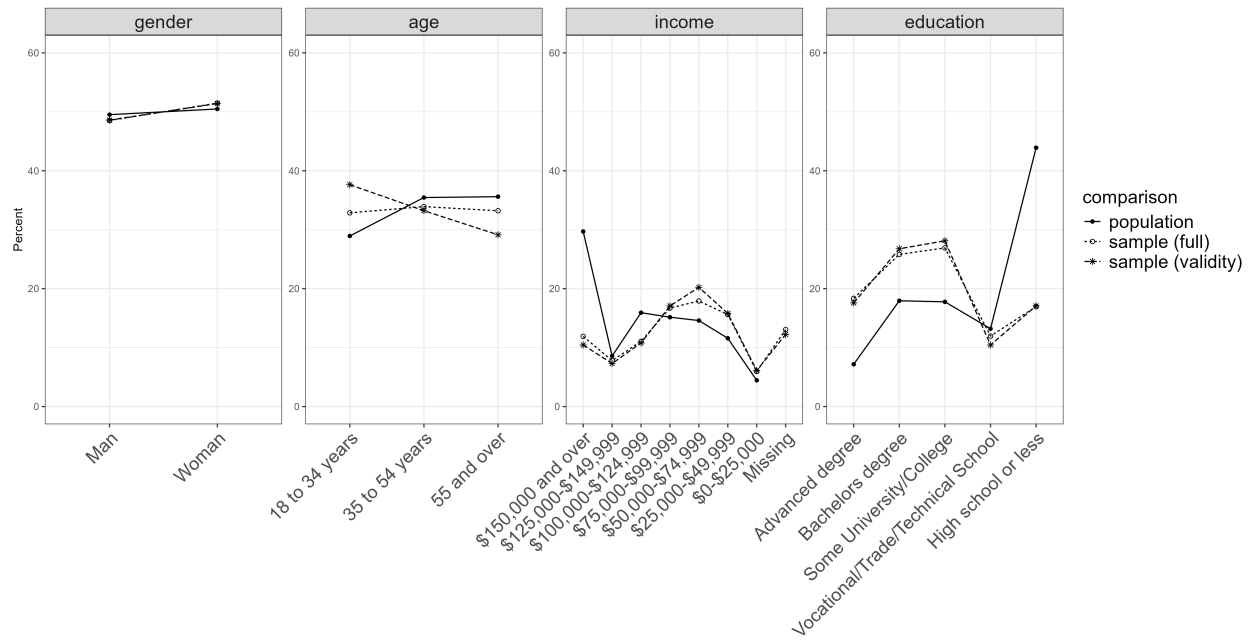


Figure A8: Comparison of wetland study sample, validity sample, and population across all three Priarie provinces

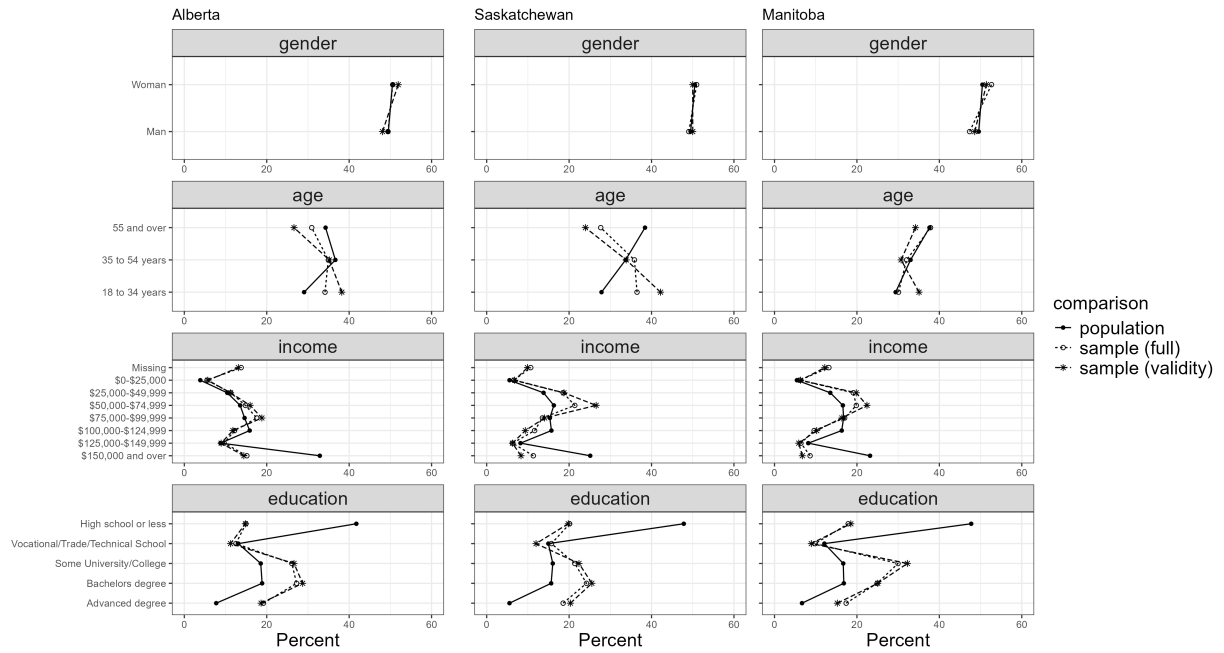


Figure A9: Comparison of wetland study sample, validity sample, and population for each province